46

Degenerative Movement Disorders of the Central Nervous System

ABU QUTUBUDDIN AND EVA ZINOVIEV

SUMMARY PEARLS

- Movement disorders are neurologic syndromes characterized by either an excess of movement (hyperkinetic) or a paucity of voluntary and autonomic movements (hypokinetic), unrelated to weakness or spasticity.
- Restless leg syndrome is characterized by deep, ill-defined discomfort or dysesthesias in the legs, which are characteristically relieved by movement.
- Tremors are rhythmic, oscillatory movements produced by alternating or synchronous contracting of antagonistic muscle pairs and can be classified as either rest tremors or action tremors.
- Parkinson disease (PD) is a neurodegenerative disorder caused by the progressive loss of dopaminergic neurons from the substantia nigra pars compacta of the midbrain in association with alpha-synuclein intracellular inclusions, termed *Lewy bodies*, with cardinal motor signs of resting tremor, bradykinesia, rigidity, and postural instability.
- The nonmotor symptoms of PD, such as cognitive deficits and depression, worsen over time and are major determinants of quality of life, nursing home placement, and progression to overall disability.
- Levodopa remains the most effective drug to treat motor symptoms of PD and is considered the gold standard treatment, but interdisciplinary team care is the most effective overall approach.

Introduction

Neurodegenerative disorders, including movement disorders, are complex multisystem disorders characterized by abnormal protein aggregates that accumulate in select regions in the central nervous system, peripheral nervous system, and autonomic nervous system.[7] Movement disorders are neurologic syndromes characterized by either an excess of movement or a paucity of voluntary and autonomic movements, unrelated to weakness or spasticity.[23] Clinically, movement disorders are manifestations of the loss of modulatory influence by the extrapyramidal system and may be classified as hyperkinetic or hypokinetic disorders. Under hyperkinetic disorders, we will discuss restless leg syndrome (RLS), tremor, dystonia, myoclonus, chorea, and tics. Under hypokinetic disorders, we will discuss Parkinson disease (PD) and Parkinson-plus syndromes, including progressive supranuclear palsy (PSP), multisystem atrophy (MSA), and corticobasal ganglionic degeneration (CBGD).

Hyperkinetic Disorders

Hyperkinetic disorders are characterized by excessive movement.[23] To characterize these disorders, one must be able to recognize the type and pattern of the involuntary movements (e.g., tremor, chorea, myoclonus). When a combination of different movements occurs in the same individual, it can be difficult to discern the correct diagnosis. It is advisable to do repeated examinations, and, if needed, video recording over time can be helpful.

Restless Leg Syndrome

RLS is characterized by deep, ill-defined discomfort or dysesthesias in the legs that arise during prolonged rest or when the patient is drowsy and trying to fall asleep, especially at night.[9] These sensory disturbances in the legs are characteristically relieved by movement. RLS may be a primary condition or occur in association with other conditions, such as diabetes mellitus, uremia, carcinoma, pregnancy, iron deficiency, or pulmonary diseases such as chronic obstructive pulmonary disease and obstructive sleep apnea.[9] Genetic studies show a hereditary component for the development of RLS, and certain genetic loci and alleles associated with the disease have been identified.[9,13,42] The pathophysiology is still poorly understood, though studies suggest that genetic susceptibility, brain iron deficiency, and deregulation of certain neurotransmitters (dopamine, glutamate, adenosine) play a role in the pathophysiology of RLS.[9] Although RLS is fairly common in the general population, its frequency, severity, and impact on sleep, mood, and quality of life vary per patient.[9]

As defined by the International Restless Leg Syndrome Study Group,[2] following are the essential diagnostic criteria:

1. An urge to move the legs usually but not always accompanied by or felt to be caused by uncomfortable and unpleasant sensations in the legs.
2. The urge to move the legs and any accompanying unpleasant sensations begin or worsen during periods of rest or inactivity, such as lying down or sitting.
3. The urge to move the legs and any accompanying unpleasant sensations are partially or totally relieved by movement, such as walking or stretching, at least as long as the activity continues.
4. The urge to move the legs and any accompanying unpleasant sensations during rest or inactivity only occur or are worse in the evening or night than during the day.
5. The occurrence of the aforementioned features is not solely accounted for as symptoms primary to another medical or behavioral condition (e.g., myalgia, venous stasis, leg edema, arthritis, leg cramps, positional discomfort, and habitual foot tapping).

Treatment of RLS includes the reduction of physical activity in the evening and decreased caffeine, alcohol, and tobacco consumption. Oral iron replacement with vitamin C is recommended in patients with low ferritin levels (<75 μg/L) and/or transferrin saturation (<20%). In patients unresponsive to oral replacement or unable to take oral medications, intravenous (IV) replacement could be considered. Medications that worsen RLS symptoms should be reduced, discontinued, or the dose should be taken in the morning. Improving sleep habits and participating in cognitive behavioral therapy can also alleviate the symptoms of RLS and insomnia.[9] According to the 2017 Movement Disorders Society evidence-based review on the treatment of RLS, ropinirole, rotigotine transdermal patch, pramipexole, cabergoline, pregabalin, gabapentin, and levodopa—each at a specific dose range—are considered efficacious for the treatment of RLS of various causes. Exercise, pneumatic compression devices, and vitamins C and E are considered likely efficacious, as are certain forms of IV iron. There is insufficient evidence to support the use of oral iron in patients who are not iron deficient. Oxycodone was determined to be efficacious for use in severe treatment-resistant RLS.[60]

Tremor

Tremors are rhythmic, oscillatory movements produced by alternating or synchronous contracting of antagonistic muscle pairs.[45] Tremors may be described as fast or slow, coarse or fine, uniplanar or biplanar. Tremors can occur in the upper and lower extremities, such as in fingers, hands, or the whole upper limb, as well as the head and parts of the face. They can occur unilaterally or symmetrically bilaterally. Tremor is usually classified as either a resting tremor or an action tremor.

Resting Tremor

Resting tremor is identified when the body part is at complete rest, as seen in PD. Resting tremor subsides with actions or with assuming a posture. PD tremor is the most common cause of resting tremor and is classically known to occur at a frequency of 4 to 6 Hz. The PD resting tremor classically develops unilaterally, often in the distal upper extremity. The term *pill-rolling tremor* refers to the development of resting tremor in the thumb and forefinger as though rolling something between the fingers. As PD progresses, it can also cause a resting tremor in the lips, jaw, chin, and legs. Other causes of rest tremor include parkinsonism and antidopaminergic medications such as antipsychotics and antiemetics. Other medications that can cause a similar tremor include carbamazepine, selective serotonin reuptake inhibitors, lithium, and amiodarone.[23,45]

Action Tremor

Action tremor occurs when muscles are voluntarily contracting and can be categorized into postural, kinetic, intention, and task-specific tremors. Action tremor includes the most common neurologic cause of tremor: essential tremor (ET). Other common action tremors include enhanced physiologic tremor, cerebellar tremor, and functional tremor.[23,45]

Essential tremor. The most common movement disorder is ET (Videos 46.1 and 46.2). Classically, ET is symmetric postural tremor with or without a kinetic component, such as a goal-directed activity, that involves the distal upper extremities. The tremor is typically uniplanar with flexion-extension movement of the hand. ET typically has a gradual onset, must have a duration of 5+ consecutive years, and cannot be explained by any other underlying disorders. There is a hereditary component to ET. More than 50% of patients inherit this disease. The tremor can be asymmetric and tend to occur more prominently in the dominant limb. It may be present in the head, voice, tongue, lips, and trunk. In severe and advanced cases, patients can develop a resting tremor. Anxiety, stress, and stimulants can worsen ET. Consumption of small quantities of alcohol can improve ET in most cases. Conservative management of ET includes occupational therapy and adaptive devices, such as weighted utensils, pens, or a computer mouse. Wearable devices on the wrist are available, which use either weight, gyroscopic forces, or peripheral nerve stimulation to stabilize the hand. The first-line pharmacotherapy includes propranolol and primidone. Second-line pharmacologic agents include topiramate, gabapentin, and benzodiazepines. In addition, botulinum toxin injections can be considered for tremors of the limb, voice, and head. Surgical options for resistant ET include deep brain stimulation (DBS) and magnetic resonance-guided focused ultrasound (FUS).[23,45,46]

Enhanced physiologic tremor. Physiologic tremor occurs because of muscle fibers whose motor units are being recruited at subtetanic rates. This occurs in all individuals with volitional muscle movement; however, the tremor is not visible because of the low amplitude of tremor. Enhanced physiologic tremor occurs with sympathetic activation. Exacerbation of physiologic tremor occurs with anxiety, fatigue, hypoglycemia, thyrotoxicosis, alcohol withdrawal, lithium, sympathomimetic drugs, caffeine, and sodium valproate. Enhanced physiologic tremor related to medication use is termed *drug-induced tremor*. It is the most common cause of postural tremor. Typically, it presents as a fine, high-frequency, low-amplitude tremor of the hands and fingers. Treatment includes removal of the offending agent. Enhanced physiologic tremor should be excluded before diagnosis of other neurologic causes of tremor.[23,45]

Cerebellar tremor. Also called rubral or coarse tremor, a cerebellar tumor classically presents as an intention tremor, defined as a tremor that worsens or only occurs when the limb is approaching a target. The tremor develops because of dysfunction of the cerebellum, which affects the precision of coordinated movements of the muscles of the limbs, seen most often in fine motor skill tasks. The tremor tends to be low frequency with high or variable amplitude. Finger-to-nose and heel-to-shin tests can elicit an intention tremor.[40]

Functional tremor.

Functional tremors are the most common functional movement disorder. They tend to have an abrupt onset, as well as inconsistency and variability in tremor frequency, amplitude, and location. Tremor amplitude tends to change when attention is brought away from the affected limb, and the tremor may be eliminated with a distraction task. The tremor may be entrainable, whereby the affected limb adopts the same tremor frequency as a tapping task performed on a different limb. Functional tremor can coexist with other functional movement disorders. The tremor is more common in females compared with males.[43]

Dystonia

Dystonia is defined as an abnormal movement characterized by sustained, painful muscle contractions, frequently causing twisting and repetitive movements, which can include prolonged abnormal postures. These movements may occur at rest or when a body part has a voluntary action. Dystonia can be initiated or worsened by voluntary action and is associated with overflow muscle activation. Dystonic movements can have a regular or irregular rhythm, though most are arrhythmic movements repeating over time at no fixed interval. The speed of dystonic movements can vary, such as the fast twitch seen in blepharospasms vs. the slower movement seen in dystonic posturing. Muscle contraction can be sustained or intermittent.[12,23]

Dystonia can also be classified according to the site of involvement:

1. Focal dystonia: One part of the body is involved, such as in blepharospasm, oromandibular dystonia, and cervical dystonia.
2. Segmental dystonia: Two or more contiguous parts are involved, such as in Meige syndrome.
3. Multifocal dystonia: Two or more noncontiguous parts are involved.
4. Hemidystonia: One side of the body is involved.
5. Generalized dystonia: Multiple muscle groups are affected throughout the body.

Dystonia can be inherited, acquired, or idiopathic. In inherited dystonia, genetic transmission can be autosomal dominant, autosomal recessive, X-linked recessive, X-linked dominant, or mitochondrial. Acquired dystonia can occur secondary to perinatal or acquired brain injury, infection, drugs, toxins, vascular, neoplastic, or psychogenic conditions. A dystonic spasm can be misdiagnosed as myoclonus; however, dystonia on electromyography studies shows prolonged bursts ($>$200 ms) rather than the short-duration bursts seen in myoclonus, which can help differentiate between the two. Dystonia may be task specific, such as writer's cramp or musician's cramp.[49] Parkinsonism can be associated with dystonia and may respond to levodopa.[47]

Patients with dystonia may report the condition is aggravated by anxiety, stress, or fatigue. Frequently, it is relieved by rest and/or sleep. Some patients are able to relieve the dystonic movements by sensory tricks (geste antagoniste), usually tactile stimuli. For example, sometimes blepharospasm can be relieved by touching the area around the eye.

Myoclonus

Myoclonus is defined as sudden, shocklike movements that are random and range from mild to severe enough to move the whole body. These movements are involuntary and cannot be suppressed. They can occur spontaneously or can be triggered by touch, light, noise, among other things. These movements can be physiologic, as seen in healthy individuals after exercise, excessive fatigue, or when falling asleep (hypnagogic jerks). Myoclonus also occurs after injury to the nervous system, either cortical or subcortical (brainstem or spinal cord), brain tumors, kidney or liver failure, lipid storage disease, chemical or drug intoxication, prolonged hypoxia, and in conjunction with several neurologic disorders, such as epilepsy, multiple sclerosis, and PD. Idiopathic myoclonus, also called essential myoclonus, develops in the absence of known abnormalities of the brain, spinal cord, or peripheral nerves. There are several subtypes of myoclonus, including action myoclonus, spinal myoclonus, palatal myoclonus, cortical reflex myoclonus, reticular reflex myoclonus, stimulus sensitive myoclonus, sleep myoclonus, and asterixis, which is a form of negative myoclonus. The first-line treatment for disruptive myoclonus is benzodiazepines.[53,57]

Chorea

The word *chorea* is derived from the Greek word *khoreia*, which means "dance." The term was introduced during the Middle Ages when chorea Sancti Viti, or dancing mania, occurred during Black Death epidemics. Chorea is defined by the irregular, involuntary, unpredictable, purposeless movements of low amplitude that flow from one body part to the next. The movements are usually distal and range in severity. These movements can be partially suppressible. Chorea results from pathologic changes in the basal ganglia. Mild chorea may resemble fidgetiness in children, whereas severe chorea may interfere with speech, swallowing, and ability to maintain posture and/or to ambulate. There are a number of diseases that can present with chorea, such as Huntington disease (HD), benign hereditary chorea, or Wilson disease.[23,51]

Huntington Disease

HD is a neurodegenerative disease characterized by chorea, gait instability, neuropsychiatric changes, and cognitive decline.[23] It is caused by a dominantly inherited cytosine-adenine-guanine trinucleotide repeat expansion in the huntingtin (HTT) gene on chromosome 4.[29] Gene testing for the mutant HTT protein can assist with the diagnosis of HD or can be used by family members to determine risk of developing HD. A systemic review of the global prevalence of HD[6] found an estimated prevalence of 5.96 to 13.7 cases per 100,000 population in North America, Northwest Europe, and Australia. The prevalence of HD in Africa and Asia is estimated to be much lower than in Western populations. There are no disease-modifying medications available at this time. There are two US Food and Drug Administration–approved medications for the treatment of HD caused by chorea: tetrabenazine and deutetrabenazine.[23]

Ballism

Ballism is a form of forceful, high-amplitude, coarse chorea, which can present after damage to the subthalamic nucleus. The movement usually presents unilaterally, involving only one-half the body, and is termed *hemiballismus*.[23]

Athetosis

Athetosis is a slow-flowing form of chorea. Athetosis is associated with perinatal brain injury, which can cause the facial grimacing and bulbar spasms seen in cerebral palsy.[23]

Tics

Tics are abnormal movements (motor tics) or abnormal sounds (phonic tics) that are brief, involuntary, rapid, and nonrhythmic. There is often an irresistible urge to move before the tic, resulting in retention that builds up and is subsequently relieved by execution of the tic. The diagnosis of Tourette syndrome is appropriate when both phonic and motor tics are present starting in early childhood.[55] Tics can be classified as simple and/or complex. Simple motor tics are abrupt, brief, isolated movements, such as an eye blink, facial grimace, shoulder shrug, or head jerk. Simple vocal tics usually consist of throat clearing, grunting, coughing, snoring, or animal sounds such as barking, hissing, and growling. Complex motor tics are patterned, coordinated movements such as touching, grooming, scratching, kicking, hand shaking, and obscene gesturing. It is sometimes difficult to distinguish complex motor tics from obsessive compulsive behavior. Complex vocal tics include words, phrases, obscene utterances, or religious profanities. Stress, anxiety, and fatigue may worsen tics. Tics can be relieved by concentrating on a task or absorbing activities such as playing a musical instrument or reading. Tics can vary in frequency, amplitude, duration, and location.[55]

Functional Tics

Functional ticlike movements are a type of functional movement disorder. These tics are phenomenologically very similar to Tourette syndrome, which makes these difficult to differentiate clinically.[17] In functional tics, onset tends to be abrupt and occurs later in the childhood/teenage years. The tics commonly interfere with voluntary movements, and there is a lack of premonitory urge. The tics tend to be nonsuppressible, and the patient often has an atypical response to anti-tic medication (lack of response or complete resolution after only one dose). Often, patients with functional ticlike movements have been exposed to social media influencers who share their experience with Tourette syndrome, suggesting that the patterns of movements displayed may serve as a model for others to unconsciously and involuntarily produce similar movements.[30]

Hypokinetic Disorders

Hypokinetic disorders are characterized by a paucity of movement, or hypokinesis.[23] This section includes PD and Parkinson-plus syndromes. Parkinson-plus syndromes are a group of neurodegenerative conditions that are characterized by the classic TRAP (tremor, rigidity, akathisia/bradykinesia, postural instability) features of PD, in addition to other features that distinguish them from PD. These conditions include PSP, MSA, and CBDG.

Parkinson Disease

Epidemiology

The prevalence of PD varies with both ethnicity and geographic distribution. Epidemiologic studies show a world prevalence between 41 and 1087 per 100,000 depending on age range.[39] PD is currently the second most common age-related neurodegenerative disease after Alzheimer disease. Experts estimate over 1 million people in the United States have PD, more than multiple sclerosis, amyotrophic lateral sclerosis, muscular dystrophy, and myasthenia gravis combined. PD has a male predominance with a male-female ratio of 3:2. The Parkinson's Foundation Prevalence Project estimates that 1.2 million people in the United States will be living with PD by 2030.[32] The prevalence of PD increases with age and is expected to increase as the baby-boomer generation ages. The Parkinson's Foundation has estimated that the combined annual direct and indirect cost of PD in the United States is $52 billion.[33]

• BOX 46.1 Top 10 Facts About Parkinson Disease

1. Named after Dr. James Parkinson (1755–1824) by the French neurologist Jean-Martin Charcot (also known as the father of neurology).
2. 1 in 100 Americans age >60 years is affected by this disease.
3. 1 in 20 people diagnosed with Parkinson disease are age <40 years.
4. Risk increases at age >50 years.
5. Parkinson disease is caused by loss of dopaminergic cells in the substantia nigra. Symptoms appear when 60%–70% of cells have died.
6. Presently, the underlying pathology causing cell death is poorly understood.
7. No cure is available, but treatments can help control symptoms and maintain quality of life.
8. Parkinson disease can be managed with a combination of drug therapies plus rehabilitation.
9. The main symptoms of Parkinson disease are tremors, bradykinesia, and rigidity.
10. The diagnosis is clinical; currently, no laboratory testing or imaging is definitive.

Pathophysiology

PD is thought to be caused by the progressive loss of dopaminergic neurons from the substantia nigra pars compacta of the midbrain in association with alpha-synuclein intracellular inclusions, termed *Lewy bodies.* Loss of these neurons is evident in the classic depigmentation of the substantia nigra of the midbrain in postmortem brain examination. It is suspected that the midbrain is affected first, resulting in subtle subclinical manifestations, but once the basal ganglia is significantly involved, classic idiopathic PD symptoms are exhibited.[1]

Advancing age is the single greatest risk factor for developing sporadic PD[10] (Box 46.1). Other pathways implicated in the pathophysiology of PD include mitochondrial dysfunction, oxidative stress, protein aggregation, impaired autophagy, and neuroinflammation.[48] The development of PD has also been linked to the use of pesticides, herbicides, and proximity to industrial plants.[34]

Another significant risk factor for developing PD is a family history of PD. Familial forms of PD account for only 10% of all PD cases. Genetic transmission can be autosomal dominant *(LRRK2, SNCA, VPS35, EIF4GI)* or autosomal recessive *(PARK2, PINK1, DJ-1)*. Additionally, there are several genes and loci that have been identified as having an association with an increased risk of developing PD.[1,16]

Clinical Presentation

The symptoms of PD can be divided into two major groupings (motor and nonmotor).

Motor Symptoms. The cardinal motor signs of PD include resting tremor, bradykinesia, rigidity, and postural instability.

Tremor. The classic early motor presentation of PD is tremor, usually unilateral and present at rest, which often begins in the distal hands and fingers. Pill-rolling tremor describes tremor that affects the thumb and forefinger, as though rolling something between them (Video 46.3).[8] The tremor may abate with actions and reemerge with rest and distraction.[8] To assess for tremor, the examiner observes the patient's hands at rest, when held at a posture, and when performing an action such as bringing the hands to the mouth or performing finger-to-nose testing. Though the tremor often presents unilaterally, as PD progresses, the tremor often begins to affect the contralateral side. In advanced PD, the patient can develop tremor with postures and actions. Tremor can begin to affect the patient's ability to perform activities of daily living (ADL), and weighted tools or external stimulation devices can be helpful nonpharmacologic options to reduce tremor.

Bradykinesia. Bradykinesia is the slowness of movements with a progressive loss of amplitude or speed during attempted rapid alternating movements of body segments.[28] This often presents as difficulty with fine motor tasks such as handwriting (micrographia, which is abnormally small, cramped handwriting). Some patients may refer to this as "weakness," though the muscle strength is normal, and this is actually the bradykinesia causing difficulty initiating a movement. Bradykinesia can also cause a patient to drag the leg on the affected side or cause shuffling gait, which is a classic symptom for PD.[1] Bradykinesia also results in "masked facies" and reduced blinking (hypomimia), lack of gestures during conversation, soft speech (hypophonia), and difficulty swallowing (dysphagia). Bradykinesia is assessed by asking the patient to perform repetitive movements as quickly and as widely as possible, such as finger taps, opening and closing the hand, and heel taps. The examiner watches the movements and observes for progressive slowness or for progressive loss of movement amplitude (decrement).[8] Physical, occupational, and sleep therapy can help mitigate the everyday challenges caused by bradykinesia.

Rigidity. Rigidity refers to increased muscle tone felt by the examiner during passive movement of the affected limb that is force dependent and velocity independent.[28] Unlike spasticity, which is velocity dependent, rigidity causes increased tone throughout the full range of motion (lead pipe rigidity). Cogwheel rigidity can be felt when the patient is affected by both rigidity and resting tremor.[8] Rigidity manifests as difficulty getting up from a chair and reduced arm swing on the affected side during ambulation. Rigidity can cause asymmetric back and shoulder pain independent of other underlying shoulder pathology.[8]

Gait and Axial Disturbances. Typical features of PD gait include a shortened slow stride (shuffling), reduced arm swing, stooped posture, and narrow base of support.[28] Patients may also have hesitancy with taking the first step or with freezing of gait.[8] It may become difficult or impossible to walk or turn, especially when asked to also perform a cognitive task (dual tasking). During ambulation, patients may have anteropulsion, where the center of gravity falls ahead of the feet causing the patient to take short fast steps to compensate for the change in center of gravity (festination).[28] The stooped posture seen in patients with PD can lead to camptocormia, a forward thoracolumbar flexion the spine greater than 45 degrees. Advanced PD can also cause lateral bending of the trunk, causing the patient to lean to one side (PISA syndrome). Postural instability, or loss of postural reflexes, leads to falls as the disease advances. Examiners can test postural instability by the pull test (clinical test that measures a patient's ability to recover balance after a sudden backward pull on the shoulders). PD patients will be unable to quickly correct and will fall back toward the examiner.[8]

Nonmotor Symptoms. Many nonmotor symptoms worsen over time and are major determinants of quality of life, nursing home placement, and progression to overall disability. These symptoms reflect the deposition of alpha-synuclein in parts of the brain other than the substantia nigra.[8] Nonmotor symptoms include psychiatric symptoms (such as apathy, anhedonia, anxiety, depression, hallucinations [mostly visual], and delusions) and cognitive impairment ranging from mild impairment to dementia.[28]

Psychiatric. Psychiatric symptoms may be underdiagnosed because of the clinical focus on motor symptoms, limited examination time, and overlap of psychiatric symptoms with other PD manifestations (e.g., masked expression, fatigue, low libido, sleep disruption). Depression may contribute to cognitive dysfunction and adds significant disability beyond that attributed to motor dysfunction alone.

Cognitive. Cognitive impairment is also frequently seen in PD and has important clinical management implications. Executive functions are the first and most severely affected of the cognitive domains.[14] This is expressed clinically as difficulties with everyday activities, such as organizing medications or paying bills. Memory is the second most frequently affected cognitive domain. Also, impairment in visuospatial function is also present early in the course of PD. The clock drawing test is a frequently used tool to assess both executive and visuospatial functions. As the disease progresses, thought processes become more rigid and preservative. In a large prospective cohort study, persons living with PD had approximately four times higher risk of developing dementia as compared to age- and sex-matched controls.[5] Various subtypes of dementia have been identified, including PD with mild cognitive impairment, PD dementia (PDD), and Lewy body dementia (LBD). Clinically and pathologically, PDD is closely related to LBD, and the two conditions are viewed as part of the alpha-synucleinopathy spectrum. The clinical distinction is drawn according to the temporal order of the onset of motor and cognitive symptoms.[35] The current convention is to term a dementia as LBD that is manifested before or within 1 year of PD symptoms, whereas PDD is diagnosed when cognitive symptoms develop at least 1 year after exhibiting PD symptoms.[21] Both share a course of progressive cognitive decline, impaired memory, and poor executive functioning. LBD also may manifest with hallucinations and delusions. Cholinesterase inhibitors and memantine are often used to treat cognitive decline.[58]

Autonomic. PD can cause dysfunction of the autonomic nervous system resulting in orthostatic hypotension (OH), constipation, urinary urgency or retention, sexual dysfunction (erectile dysfunction and atrophic vaginitis), excessive sweating, dry eyes, and sialorrhea.[28] OH most commonly occurs in the advanced stages of PD.

Sleep. Sleep disturbances are common in PD, although their cause is unclear and likely multifactorial. PD can cause sleep disorders such as insomnia, RLS, periodic limb movements in sleep, excessive daytime sleepiness, and rapid eye movement (REM) sleep disorder.[28] Factors that may contribute to sleep disturbances in PD include stimulant effects of PD medication, anxiety, depression, sleep cycle dysregulation, reduced bed mobility, frequent nocturia, cognitive impairment, and vivid dreams. REM sleep behavior disorder may be an early presenting sign in some PD patients.[41a]

Special Senses. PD can also affect some sensory processes such as hyposmia (reduced sense of smell), impaired visual acuity and contrast perception, poor visual tracking and motion perception, and dysesthesias/paresthesias[28] (Box 46.2).

Diagnosis

Understanding the clinical presentation of PD is especially important because the diagnosis of PD is largely based on clinical assessment. The most widely used comprehensive instrument used by movement disorder specialists to assess the symptoms of PD is the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS; revised in 2008).[19] The UPDRS is a clinical rating scale consisting of four parts: nonmotor experiences in daily living, motor experiences of daily living, motor examination, and motor complications. Parts of

• BOX 46.2 Top 10 Warning Signs for Parkinson Disease

1. Difficulty walking (difficulty starting and stopping, misjudging corners, abnormal shoe wear)
2. Resting tremors (usually unilateral at onset)
3. Bradykinesia
4. Difficulty balancing (carrying items while walking becomes difficult, frequent falls)
5. Depression/sadness (often precedes onset of motor signs)
6. Loss of motor skills
7. Handwriting (micrographia, tremulousness)
8. Voice and speech changes/difficulty (hypophonia, dysarthria)
9. Loss of memory/dementia (mild or severe)
10. Skin disorders (dry skin, rough skin, dandruff of scalp)

the scale can be completed by the patient and/or caregiver via a questionnaire, and other parts require the assessment of a movement disorders specialist.

Prior to providing a diagnosis of PD, it is important to exclude disease processes that may mimic PD.[28] Initially, blood tests can help rule out other medical conditions that can cause slowness, such as hypothyroidism or anemia. It is also crucial to rule out drug-induced parkinsonism by reviewing the patient's medication list. Typically, the culprits are antipsychotics, metoclopramide, amiodarone, lithium, or reserpine. These medications should be reduced or discontinued, if possible. Patients with drug-induced parkinsonism rarely develop certain nonmotor manifestations, such as anosmia, and the dopamine transporter (DAT) scan will be negative. Toxin-induced parkinsonism seen in MPTP (1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine), manganese, or carbon monoxide toxicity requires removal of the toxin. Vascular parkinsonism, or multiinfarct parkinsonism, is a condition in which PD-like symptoms develop as a result of cerebrovascular disease, causing small lacunar infarcts in the basal ganglia.[54] It usually presents with symmetric onset of symptoms, without tremor but the with the presence of corticospinal tract signs. It is typically unresponsive to levodopa, though some studies show a positive or sufficient response to levodopa.

Magnetic resonance imaging (MRI) or computed tomography scan can be useful in narrowing the differential when the clinical presentation is unclear.[28] Brain imaging can rule out other neurologic conditions, such as normal pressure hydrocephalus or subcortical stroke. The DAT scan can be used to detect presynaptic dopaminergic neuron degeneration in the basal ganglia. The DAT scan can be considered in rare instances when the diagnosis is unclear, and the differential includes both ET and PD. Recently, alpha-synuclein skin biopsy (Syn-One skin test) and cerebral spinal fluid testing (seed amplification assay) have become more readily available for clinical use.[31] Studies have found these tests are sensitive and specific for diagnosing alpha-synucleinopathies.[18] These tests cannot differentiate PD from other neurodegenerative diseases characterized by alpha-synuclein deposition, such as LBD and MSA. Infrequently, genetic testing can be used to aid in PD diagnosis.

The MDS has created the Clinical Diagnostic Criteria for Parkinson's Disease to help diagnose PD.[37] This tool guides the evaluation of clinical symptoms of PD, defines exclusion criteria and red flags that may suggest another disease process, and outlines supportive criteria to assist in the diagnosis of PD.

Management

The International Parkinson and Movement Disorder Society Evidence-Based Medicine Committee (MDS-EBM) published updates regarding its recommendations for symptomatic treatment of PD in 2018.[15] The publication provides clinicians with updated guidelines to guide pharmacologic and nonpharmacologic treatment of PD.

Pharmacologic Treatment of Motor Symptoms. Individual assessment of the risks and benefits of available medications, as well as specific clinical features and phases of the disease, should guide PD treatment.[38] Pharmacologic treatment becomes necessary when PD-related motor symptoms begin to interfere with the patient's ADL and negatively affect quality of life. Levodopa remains the most effective drug to treat motor symptoms of PD and is considered the gold standard. It is recommended as a first-line initial monotherapy in elderly patients in whom motor complication risk is generally low.[38] For younger patients, alternative agents are often used as first-line therapy because of the risk of development of levodopa-related motor complications, such as motor response oscillations or levodopa-induced dyskinesias (LIDs).

When initiating therapy, providers should prescribe the lowest effective dose of levodopa.[38] Patients should be monitored for the development of dyskinesias, nausea, impulse control disorders, hallucinations, excessive daytime sleepiness, and worsening of postural orthostatic hypotension while on levodopa. Nausea is often an initial side effect of levodopa and can be ameliorated by taking the medication with food. However, patients should be counseled that taking levodopa with a meal high in protein can reduce the absorption and efficacy of the medication.

As PD disease progresses, adjunct medications can be considered to optimize management. In addition to serving as adjuncts to levodopa treatment, catechol-O-methyltransferase (COMT) inhibitors, monoamine oxidase B (MAO-B) inhibitors, and dopamine agonists (DAs) may be used as monotherapy. Advantages of these agents include a more continuous oral delivery of dopaminergic stimulation and a lower risk of levodopa-induced motor complications.[38] Box 46.3 lists the pharmacologic treatment options for PD by class.

Motor Response Oscillations (The On-Off Phenomenon). Because of the short half-life of levodopa, patients may experience motor response oscillations, most frequently occurring in the form of wearing-off phenomenon.[4] Patients will describe worsening symptoms toward the end of the intradose interval, with relief after intake of the next scheduled levodopa dose. COMT inhibitors, MOA-B inhibitors and DAs prolong the half-life of levodopa and therefore help to control motor fluctuations and help to reduce total daily time off.[4]

Levodopa-Induced Dyskinesias. LIDs are characterized by choreatic movements of the extremities, trunk, and neck during periods of full efficacy and are therefore considered on state dyskinesias (Video 46.4). Around 80% of PD patients develop LID, with 30% developing it after just 3 years of levodopa treatments.[25] An option to reduce LIDs includes decreasing the dose of levodopa, although this may improve LIDs at the expense of symptom control. Amantadine, an N-methyl-D-aspartate receptor antagonist, is widely recognized as the most effective drug to manage LID.[25] Studies have shown that amantadine can decrease LIDs without worsening parkinsonian symptoms and that the antidyskinetic effects can be maintained long term. Adverse effects of amantadine include the potential to induce cognitive dysfunction, including psychosis. Therefore

• BOX 46.3 Pharmacologic Treatment Options for Parkinson Disease

Levodopa-Based Treatment Options (With Carbidopa)
Immediate-release tablet (Sinemet)
Extended-release tablet (Sinemet CR; Rytary)
Orally disintegrating tablets (Parcopa)
Enteral suspension (Duopa)

Levodopa-Based Treatment Options (Without Carbidopa)
Capsule for inhalation (Inbrija)

Dopamine Agonists
Nonergot preparations
  Pramipexole tablet (Mirapex)
  Ropinirole tablet (Requip)
  Rotigotine transdermal patch (NeuPro)
  Apomorphine subcutaneous injection (Apokyn)
Ergot preparations
  Bromocriptine tablet (Parlodel)
  Pergolide tablet (Prascend)

COMT Inhibitors
Entacapone tablet (Comtan)
Carbidopa/levodopa/entacapone tablet (Stalevo)

MAO Inhibitors (Less Common)
Safinamide tablet (Sadako)
Selegiline capsule, tablet (Eldepryl)
Selegiline orally disintegrating tablet (Zelapar)
Selegiline transdermal patch (Emsam)
Rasagiline tablet (Azilect)

NMDA Receptor Antagonist
Amantadine capsule, extended-release capsule, tablet, extended-release tablet

Adenosine A2A Receptor Inhibitor
Istradefylline tablet (Nourianz)

Anticholinergic Agents
Benztropine tablet (Cogentin)
Trihexyphenidyl tablet, oral solution (Artane)

Selective Serotonin Inverse Antagonist
Pimavanserin capsule, tablet (Nuplazid)

COMT, Catechol-O-methyltransferase; *MAO,* monoamine oxidase; *NMDA,* N-methyl-D-aspartate.

caution should be used when giving this medication to patients with PD dementia.

Surgical and Device-Aided Treatments. DBS is a surgical procedure used as part of a comprehensive treatment approach to managing motor symptoms in PD (Video 46.5). In DBS for PD, an electrode is surgically implanted in the subthalamic nucleus (STN) or globus pallidus interna (GPi) and provides continuous high-frequency electrical stimulation.[4] DBS is usually indicated for patients with disabling drug-related movements and motor fluctuations whose condition has not improved after exhausting all medical management regimens, who show a clear response to levodopa after a levodopa challenge, who do not have a parkinsonism syndrome, and who have no other major interfering medical conditions. The ventralis intermedius nucleus of the thalamus is another potential lead target in PD patients with severe tremor who may not be candidates for lead placement at the STN or GPi because of dementia or other major medical conditions. Surgical DBS lead placement was once performed using a stereotactic frame; however, neurosurgeons at certain major surgical centers are now able to place leads via frameless stereotaxy.[36] The DBS leads are attached to a battery that is placed in the chest wall similar to a pacemaker. The device can then be accessed externally via a programmer to fine-tune stimulation to optimize motor symptom treatment.

Another available procedure is MRI FUS, the noninvasive procedure that causes unilateral ablation of the thalamus for treatment of severe tremor[4] or of the STN for treatment of medication-resistant PD motor symptoms.[27] Other device-aided treatments used in advanced PD include levodopa-carbidopa (Duopa) intestinal gel infusion, subcutaneous injection of dopamine agonist (apomorphine), and an inhaled levodopa powder (Inbrija) given via an inhaler for treatment of severe off episodes.[38]

Nonpharmacologic Treatment. The management of PD requires an interdisciplinary approach. The MDS-EBM Review of Treatment update in 2018[15] concluded that physical therapy is likely efficacious for the treatment of PD motor symptoms (Table 46.1). After review of the available studies, the update suggests that physical therapy interventions are clinically useful in the treatment of PD. There was insufficient evidence for the efficacy of exercise-based movement training strategies, though the implication for clinical practice was deemed possibly useful. Additionally, there was insufficient evidence for the efficacy of technology-based movement strategy training with the results of the review suggesting the implication for clinical practice remains investigational. There is insufficient evidence for the efficacy of formalized patterned exercise such as tai chi or dance in the treatment of PD, and the implication for clinical practice is possibly useful. Currently, there is also insufficient evidence for the efficacy of occupational therapy and video-assisted swallowing therapy in the treatment of PD, though the clinical implications are possibly useful.

Caregiver Considerations. The patient with a movement disorder frequently becomes increasingly reliant on a caregiver to assist with a multitude of daily tasks. Caregivers (either unpaid or paid) are an intrinsic part of the patient's life and can serve a useful role in the medical treatment process.[26] At home, caregivers may help the patient with a wide array of tasks that include assisting with ADLs, increasing safety awareness, medication management, general organizational issues, transportation, and social involvement. At the medical appointment, caregivers can serve as accurate historians, firsthand observers of the patient's reactions to medications and treatments, and can help clarify patient communication to medical personnel, and vice versa. For these reasons, supporting the functioning of caregivers is vital to the successful medical management and maintenance of a home/community living environment in individuals with movement disorders. Physicians and medical team members need to be alert for signs of undue caregiver burden or strain and intervene for the welfare of the caregiver and movement disorder patient alike. Interventions may take the form of simple reassurance, caregiver education, prescribing of respite breaks, or end-of-life discussions.

TABLE 46.1 Conclusions From the Movement Disorder Society Evidence-Based Medicine Committee Review of Treatment 2019 Regarding the Efficacy of Treatments for Nonmotor Symptoms of Parkinson Disease

Nonmotor Symptom	Medication	Efficacy Conclusion	Practice Implication
Depression	Pramipexole	Efficacious	Clinically useful
	Venlafaxine	Efficacious	Clinically useful
	Nortriptyline	Likely efficacious	Possibly useful
	Desipramine	Likely efficacious	Possibly useful
	Cognitive Behavioral Therapy (CBT)	Likely efficacious	Possibly useful
Apathy	Rivastigmine	Efficacious	Possibly useful
Psychosis	Clozapine	Efficacious	Clinically useful
	Pimavanserin	Efficacious	Clinically useful
Dementia	Rivastigmine	Efficacious	Clinically useful
Orthostatic hypotension	Droxidopa	Efficacious (short term)	Possibly useful
Sialorrhea	Botulinum toxin A/B	Efficacious	Clinically useful
	Glycopyrrolate	Efficacious	Possibly useful

From Seppi K, Ray Chaudhuri K, Coelho M, et al. Update on treatments for nonmotor symptoms of Parkinson's disease—an evidence-based medicine review. *Move Dis.* 2019;34(2):180-198.

Parkinson-Plus Syndromes

Progressive Supranuclear Palsy

PSP is an adult-onset, rapidly progressive neurodegenerative syndrome characterized by axial rigidity, bradykinesia, postural instability with falls, cognitive deficits, and supranuclear vertical gaze palsy.[22] PSP is a common cause of levodopa-nonresponsive parkinsonism. A systematic review in 2022 reported a pooled prevalence rate of 7.1 per 100,000.[52] Incidence increases sharply with age, with no convincing cases reported in literature less than age 40 years.[22] The characteristic histopathologic feature of PSP is an intracerebral aggregation of hyperphosphorylated tau protein as neurofibrillary tangles.[41] There is increasing evidence that mitochondrial dysfunction plays a role in the pathogenesis of neurodegenerative disease. In PSP, failure in mitochondrial energy production is probably involved in both tau aggregation and neuronal cell death.[56]

The most common initial feature of PSP is a disturbance of gait and history of falling, which typically begins within 12 months of onset. Other clinical clues indicative of PSP include stuttering, palilalia (abnormal repetition of syllables, words, or phrases), early dysphagia, personality changes, sleep disturbances, apraxia, and dementia. The characteristic facial expression of PSP is that of perpetual astonishment caused by continuous frontalis contraction and a low blink rate. Extensor neck posturing may also be seen. Other clinical findings of PSP include action tremor, pseudobulbar palsy, hyperreflexia, and Babinski signs. The MDS has published clinical diagnostic criteria for PSP for use in research and clinical practice.[22] "Definite PSP" is diagnosed through neuropathologic evaluation postmortem. "Probable PSP" can be diagnosed in a patient with either supranuclear gaze palsy *or* slow velocity of vertical saccades *and* repeated falls *or* tendency to fall *or* progressive gait freezing *or* features of parkinsonism *or* frontal/cognitive behavioral changes. The criteria further describe "possible PSP" and "suggestive of PSP." However, diagnosis of PSP may be difficult because of overlap in signs and symptoms with PD and MSA. The classic syndrome of PSP differs from PD in the lack of tremor, the relative sparing of limb movement (except writing), the greater rigidity seen in the neck as compared with the limbs, and early development of freezing gait. However, resting tremor, urinary incontinence, hemidystonia, and asymmetric apraxia are infrequently seen in PSP, though the presence of these signs and symptoms should not rule out the diagnosis of PSP. Furthermore, although supranuclear vertical gaze palsy is the most distinctive clinical feature of PSP, this finding may be absent. Therefore strict reliance on the single physical sign leads to lack of sensitivity or at least a delay in diagnosis. In addition, autonomic dysfunction may be more common than is generally appreciated, resulting in missed diagnosis of PSP as MSA.

Multiple System Atrophy

MSA is an adult-onset, rare, rapidly progressive, and fatal neurodegenerative disorder characterized by autonomic failure, poor levodopa-responsive parkinsonism, cerebellar ataxia, and pyramidal signs.[50] MSA is characterized by an abnormal accumulation of alpha-synuclein in oligodendrocytes associated with multifocal neuronal degeneration. The estimated crude prevalence of MSA ranged from 0.52 to 17 per 100,000 persons and was higher among males than females.[24] The mean age of onset for MSA is approximately 56 years,[20] which is slightly younger than that for PD and considerably younger than that for PSP.

MSA is characterized clinically by a variable combination of parkinsonism, cerebellar dysfunction, and autonomic failure. The clinical manifestations of MSA are often divided into motor and autonomic manifestations.[50] The motor syndromes in MSA can be subdivided into MSA-P and MSA-C based on the predominant motor feature. MSA-P is a basal ganglia type, characterized by levodopa-refractory, akinetic-rigid parkinsonism. Signs include bradykinesia, rigidity, postural tremor, disequilibrium, and unsteady gait. MSA-C is a cerebellar type, characterized by ataxia. Other cerebellar findings include gait ataxia, limb kinetic ataxia, wide-based gait, dysarthric and scanning speech, and cerebellar oculomotor disturbances, such as nystagmus. The most common presentation of MSA is MSA-P.

Autonomic dysfunction in MSA manifests as orthostatic lightheadedness, syncope, urinary retention or incontinence, impotence in males or anorgasmia in females, fecal incontinence, loss of sweating, and paroxysmal bursts of excessive sweating. Retrospective analysis of pathologically confirmed MSA cases showed that early autonomic symptoms (within 2 years of disease onset) occurred in more than one-half of cases. Other clinical features of MSA may include action myoclonus, dysphonia, Raynaud phenomenon, pain, contractures, and REM sleep behavior disorder, which may proceed motor manifestations. A consensus committee published formal criteria for the clinical diagnosis of MSA.[59] The central elements of diagnoses were felt to be a combination of autonomic dysfunction and either cerebellar ataxia or parkinsonism poorly responsive to levodopa. The clinical recognition of MSA improved dramatically after the introduction of diagnostic criteria. However, there is no definitive diagnostic test for MSA. Clinically probable MS is defined as a sporadic, progressive, adult-onset (age >30 years) disease characterized by severe autonomic failure as defined by either urinary incontinence (with erectile dysfunction in males) or OH (with blood pressure falls >30 mm Hg systolic or >15 mm Hg diastolic) and a poorly levodopa-responsive parkinsonism or cerebellar syndrome. Of note, MSA has emerged as the most difficult disease to distinguish from PD. MSA mimics PD more closely than PSP and the prominence of limb involvement, which is often asymmetric. Postural stability is compromised early in both MSA and PSP, but in contrast to PSP, recurrent falls at disease onset are unusual in MSA.

MSA is a fatal disorder and progresses relentlessly until death with a median survival of 9 years. However, there is considerable variation in disease progression, with individual cases surviving up to 15 years. Thus far, disease-modifying medical treatments are lacking despite intensified efforts to develop interventional strategies; therefore management of MSA patients largely concerns the alleviation of parkinsonism and autonomic symptoms.[20] Only one-third of MSA patients show any response to levodopa, and then the response is almost always atypical. If levodopa is unsuccessful or poorly tolerated, it is worth pursuing treatment with DAs or amantadine because individual patients may respond better to one of these agents. There is no consistently effective treatment for cerebellar manifestations. Autonomic manifestations are most amenable to treatment, and a variety of measures are effective for OH. A priority is to identify and reduce or eliminate medications that may be contributing to the problem. Although autonomic failure is almost universally present in MSA, only one-third of affected patients receive appropriate pharmacologic treatment. Nonpharmacologic measures include elevating the head of the bed, increasing salt and fluid intake, avoiding heat, and adopting specific body postures such as crossing legs or squatting.

Corticobasal Ganglionic Degeneration

CBGD is a neurodegenerative disease characterized by tau protein deposition and neuronal cell loss.[11] The distribution of tau pathology and gliosis determines the clinical presentation. CBGD develops insidiously and progresses gradually. CBGD is quite rare, and accurate epidemiologic studies are lacking. The clinical findings of CBGD can be divided into three categories: motor manifestations, cerebellar manifestations, and other manifestations. Motor manifestations include dystonia (usually fixed and often causing pronounced and/or painful deformities), postural instability, athetosis, and orofacial dyskinesias. Signs of cerebrocortical dysfunction consists of apraxia, cortical sensory loss, alien limb phenomenon, dementia, and frontal lobe reflexes. The most striking aspect of the clinical picture of CBGD is the asymmetry with which it presents and pursues its course.[11]

A helpful diagnostic tool is brain imaging by MRI. This will detect asymmetric atrophy of the cerebral cortex,[11] which will be greater contralateral to the side that is more clinically involved and can help exclude other pathophysiologic processes. A committee of neurologists, neuropsychologists, and movement disorders specialists determined diagnostic criteria in 2013, which defines “probable CBGD,” “possible CBGD,” and several specific phenotypes of the disease.[3] There is no specific treatment for CBGD. Therapy is currently purely symptomatic. The mean life expectancy from onset is 6.6 years.[3]

Summary

Movement disorders present a significant challenge to the diagnostic and management skills of physiatrists and are common enough to be likely encountered by physiatrists in all types of clinical settings (Table 46.2). There is still a lack of evidence-based and patient-specific rehabilitative interventions for individuals diagnosed with movement disorders. Anecdotal studies focusing on different rehabilitative exercise programs have suggested that the quality of life for movement disorder patients may be significantly improved by participation in such programs.

TABLE 46.2 Symptom Manifestation of Various Movement Disorders

Disorder	Resting Tremor	Action Tremor	Rigidity	Postural Instability	Cognitive Decline	Sleep Disorder	Visual Hallucinations	Positive Response to PD Medications	Mood Disorders
PD	++	±	+	+	+	+	±	+++	++
ET	±	++	−	−	−	−	−	+	±
MSA	−	+	+	+	++	+	±	−	−
PSP	±	−	±	+++	++	+	±	−	−
HD	±	±	−	++	++	+	±	−	±
RLS	−	−	−	−	−	+	−	±	−

+, Commonly present; −, usually absent; ±, may or may not be present.
ET, Essential tremor; *HD,* Huntington disease; *MSA,* multiple system atrophy; *PD,* Parkinson disease; *PSP,* progressive supranuclear palsy; *RLS,* restless leg syndrome.

References

1. Albin RL. *Parkinson disease*. Online ed. Oxford Academic; 2022.
2. Allen RP, Picchietti DL, Garcia-Borreguero D, et al. Restless legs syndrome/Willis-Ekbom disease diagnostic criteria: updated International Restless Legs Syndrome Study Group (IRLSSG) consensus criteria—history, rationale, description, and significance. *Sleep Med*. 2014;15(8):860-873.
3. Armstrong MJ, Litvan I, Lang AE, et al. Criteria for the diagnosis of corticobasal degeneration. *Neurology*. 2013;80(5):496-503.
4. Armstrong MJ, Okun MS. Diagnosis and treatment of Parkinson disease: a review. *JAMA*. 2020;323(6):548-560.
5. Åström DO, Simonsen J, Raket LL, et al. High risk of developing dementia in Parkinson's disease: a Swedish registry-based study. *Sci Rep*. 2022;12(1):16759.
6. Baig SS, Strong M, Quarrell OW. The global prevalence of Huntington's disease: a systematic review and discussion. *Neurodegen Dis Manage*. 2016;6(4):331-343.
7. Basha S, Mukunda DC, Rodrigues J, et al. A comprehensive review of protein misfolding disorders, underlying mechanism, clinical diagnosis, and therapeutic strategies. *Age Res Rev*. 2023;90:102017.
8. Bhatia KP, Chaudhuri KR, Stamelou M. *Parkinson's Disease*. Academic Press; 2017:vol. 132.
9. Chenini S, Barateau L, Dauvilliers Y. Restless legs syndrome: from clinic to personalized medicine. *Rev Neurol*. 2023;179(7):703-714.
10. Coleman C, Martin I. Unraveling Parkinson's disease neurodegeneration: does aging hold the clues? *J Parkinsons Dis*. 2022;12(8):2321-2338.
11. Constantinides VC, Paraskevas GP, Paraskevas PG, et al. Corticobasal degeneration and corticobasal syndrome: a review. *Clin Park Relat Disord*. 2019;1:66-71.
12. di Biase L, Di Santo A, Caminiti ML, et al. Classification of dystonia. *Life (Basel)*. 2022;12(2):206.
13. Didriksen M, Nawaz MS, Dowsett J, et al. Large genome-wide association study identifies three novel risk variants for restless legs syndrome. *Commun Biol*. 2020;3:703.
14. Fengler S, Liepelt-Scarfone I, Brockmann K, et al. Cognitive changes in prodromal Parkinson's disease: a review. *Move Dis*. 2017;32(12):1655-1666.
15. Fox SH, Katzenschlager R, Lim SY, et al. International Parkinson and Movement Disorder Society evidence-based medicine review: update on treatments for the motor symptoms of Parkinson's disease. *Move Dis*. 2018;33(8):1248-1266.
16. Funayama M, Nishioka K, Li Y, et al. Molecular genetics of Parkinson's disease: contributions and global trends. *J Hum Genet*. 2023;68(3):125-130.
17. Ganos C, Martino D, Espay AJ, et al. Tics and functional tic-like movements: can we tell them apart? *Neurology*. 2019;93(17):750-758.
18. Gibbons CH, Garcia J, Wang N, et al. The diagnostic discrimination of cutaneous α-synuclein deposition in Parkinson disease. *Neurology*. 2016;87(5):505-512.
19. Goetz CG, Tilley BC, Shaftman SR, et al. Movement Disorder Society–sponsored revision of the Unified Parkinson's Disease Rating Scale (MDS-UPDRS): scale presentation and clinimetric testing results. *Move Dis*. 2008;23(15):2129-2170.
20. Goh YY, Saunders E, Pavey S, et al. Multiple system atrophy. *Pract Neurol*. 2023;23(3):208-221.
21. Gomperts SN. Lewy body dementias: dementia with Lewy bodies and Parkinson disease dementia. *Continuum (Minneap Minn)*. 2016;22(2):435-463.
22. Hoglinger GU, Respondek G, Stamelou M, et al. Clinical diagnosis of progressive supranuclear palsy: the Movement Disorder Society criteria. *Mov Disord*. 2017;32(6):853-864.
23. Jankovic J, Hallett M, Okun MS, et al. *Principles and Practice of Movement Disorders*. 3rd ed. Elsevier; 2022:iii.
24. Kaplan S. Prevalence of multiple system atrophy: a literature review. *Rev Neurol*. 2024;180(5):438-450.
25. Kwon DK, Kwatra M, Wang J, et al. Levodopa-induced dyskinesia in Parkinson's disease: pathogenesis and emerging treatment strategies. *Cells*. 2022;11(23):3736.
26. Macchi ZA, Koljack CE, Miyasaki JM, et al. Patient and caregiver characteristics associated with caregiver burden in Parkinson's disease: a palliative care approach. *Ann Palliat Med*. 2020;9(1):S24-S33.
27. Martínez-Fernández R, Natera-Villalba E, Máñez Miró JU, et al. Prospective long-term follow-up of focused ultrasound unilateral subthalamotomy for Parkinson disease. *Neurology*. 2023;100(13):e1395-e1405.
28. Massano J, Bhatia KP. Clinical approach to Parkinson's disease: features, diagnosis, and principles of management. *Cold Spring Harb Perspect Med*. 2012;2(6):a008870.
29. McColgan P, Tabrizi SJ. Huntington's disease: a clinical review. *Eur J Neurol*. 2018;25(1):24-34.
30. Müller-Vahl KR, Edwards MJ. Mind the difference between primary tics and functional tic-like behaviors. *Move Dis*. 2021;36(12):2716-2718.
31. Orrù CD, Ma TC, Hughson AG, et al. A rapid α-synuclein seed assay of Parkinson's disease CSF panel shows high diagnostic accuracy. *Ann Clin Transl Neurol*. 2021;8(2):374-384.
32. National Institute of Neurological Disorders and Stroke. *Parkinson's Disease: Challenges*, Progress, and Promise. Accessed July 6, 2024. https://www.ninds.nih.gov/current-research/focus-disorders/parkinsons-disease-research/parkinsons-disease-challenges-progress-and-promise
33. Parkinson's Foundation. *Statistics*. Accessed July 6, 2024. https://www.parkinson.org/understanding-parkinsons/statistics.
34. Paul KC, Sinsheimer JS, Rhodes SL, et al. Organophosphate pesticide exposures, nitric oxide synthase gene variants, and gene-pesticide interactions in a case-control study of Parkinson's disease, California (USA). *Environ Health Perspect*. 2016;124(5):570-577.
35. Pennington C, Duncan G, Ritchie C. Altered awareness of cognitive and neuropsychiatric symptoms in Parkinson's disease and dementia with Lewy bodies: a systematic review. *Int J Geriatr Psychiatry*. 2021;36(1):15-30.
36. Piano C, Bove F, Mulas D, et al. Frameless stereotaxy in subthalamic deep brain stimulation: 3-year clinical outcome. *Neurol Sci*. 2021;42(1):259-266.
37. Postuma RB, Berg D, Stern M, et al. MDS clinical diagnostic criteria for Parkinson's disease. *Move Dis*. 2015;30(12):1591-1601.
38. Pringsheim T, Day GS, Smith DB, et al. Dopaminergic therapy for motor symptoms in early Parkinson disease practice guideline summary: a report of the AAN Guideline Subcommittee. *Neurology*. 2021;97(20):942-957.
39. Pringsheim T, Jette N, Frolkis A, et al. The prevalence of Parkinson's disease: a systematic review and meta-analysis. *Move Dis*. 2014;29(13):1583-1590.
40. Rocha Cabrero F, De Jesus O. *Intention Tremor*. StatPearls Publishing; 2024.
41. Rowe JB, Holland N, Rittman T. Progressive supranuclear palsy: diagnosis and management. *Pract Neurol*. 2021;21(5):376-383.

41a. Schenck CH, Bundlie SR, Mahowald MW. Delayed emergence of a parkinsonian disorder in 38% of 29 older men initially diagnosed with idiopathic rapid eye movement sleep behavior disorder. *Neurology*. 1996;46(2):388-393.

42. Schormair B, Zhao C, Bell S, et al. Identification of novel risk loci for restless legs syndrome in genome-wide association studies in individuals of European ancestry: a meta-analysis. *Lancet Neurol*. 2017;16(11):898-907.
43. Schwingenschuh P, Espay AJ. Functional tremor. *J Neurol Sci*. 2022;435:120208.
44. Seppi K, Ray Chaudhuri K, Coelho M, et al. Update on treatments for nonmotor symptoms of Parkinson's disease—an evidence-based medicine review. *Move Dis*. 2019;34(2):180-198.
45. Sharma S, Pandey S. Approach to a tremor patient. *Ann Indian Acad Neurol*. 2016;19(4):433-443.
46. Sharma S, Pandey S. Treatment of essential tremor: current status. *Postgrad Med J*. 2020;96(1132):84.

47. Shetty AS, Bhatia KP, Lang AE. Dystonia and Parkinson's disease: what is the relationship? *Neurobiol Dis*. 2019;132:104462.
48. Simon DK, Tanner CM, Brundin P. Parkinson disease epidemiology, pathology, genetics, and pathophysiology. *Clin Geriatr Med*. 2020;36(1):1-12.
49. Stahl CM, Frucht SJ. Focal task specific dystonia: a review and update. *J Neurol*. 2017;264(7):1536-1541.
50. Stankovic I, Fanciulli A, Sidoroff V, et al. A review on the clinical diagnosis of multiple system atrophy. *Cerebellum*. 2023;22(5):825-839.
51. Stimming EF, Bega D. Chorea. *Continuum (Minneap Minn)*. 2022; 28(5):1379-1408.
52. Swallow DMA, Zheng CS, Counsell CE. Systematic review of prevalence studies of progressive supranuclear palsy and corticobasal syndrome. *Move Dis Clin Pract*. 2022;9(5):604-613.
53. US Department of Health and Human Services. *Myoclonus*. National Institutes of Health; 2012.
54. Udagedara TB, Dhananjalee Alahakoon AM, et al. Vascular parkinsonism: a review on management updates. *Ann Indian Acad Neurol*. 2019;22(1):17-20.
55. Ueda K, Black KJ. A comprehensive review of tic disorders in children. *J Clin Med*. 2021;10(11):2479.
56. Valentino RR, Tamvaka N, Heckman MG, et al. Associations of mitochondrial genomic variation with corticobasal degeneration, progressive supranuclear palsy, and neuropathological tau measures. *Acta Neuropathol Commun*. 2020;8:162.
57. van der Veen S, Caviness JN, Dreissen YEM, et al. Myoclonus and other jerky movement disorders. *Clin Neurophysiol Pract*. 2022;7: 285-316.
58. Watts KE, Storr NJ, Barr PG, et al. Systematic review of pharmacological interventions for people with Lewy body dementia. *Aging Mental Health*. 2023;27(2):203-216.
59. Wenning GK, Stankovic I, Vignatelli L, et al. The Movement Disorder Society criteria for the diagnosis of multiple system atrophy. *Move Dis*. 2022;37(6):1131-1148.
60. Winkelmann J, Allen RP, Högl B, et al. Treatment of restless legs syndrome: evidence-based review and implications for clinical practice (revised 2017). *Move Dis*. 2018;33(7):1077-1091.